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SYSTEM AND METHOD FOR ROUTING OPTICAL FIBERS WITHIN A MULTIPLE DWELLING UNIT

Abstract

The present disclosure relates to a system for routing optical fibers within multiple dwelling units (MDUs) where the system has a removable optical fiber feature that enables such routing of optical fibers within the MDUs.

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Background/Summary

PRIORITY APPLICATION [0001] This application claims the benefit of priority of U.S. Provisional Application No. 63/559,392, filed on Feb. 29, 2024, the content of which is relied upon

and incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present disclosure is directed to a system for routing optical fibers within a multi-dwelling unit.

BACKGROUND OF THE INVENTION

[0003] Several hundred million multiple dwelling units (MDUs) exist globally, which are inhabited by about one third of the world's population. Due to the large concentration of tenants in one MDU, Fiber-to-the-X ("FTTX") deployments to these structures are more cost effective to service providers than deployments to single-family homes. Connecting existing MDUs to the FTTX network can often be difficult. Challenges can include gaining building access, limited distribution space in riser closets, and space for cable routing and management. Specifically, FTTX deployments within existing structures make it difficult to route cables within the walls or floors, or above the ceiling from a central closet or stairwell, to each living unit.

[0004] Conventionally, a service provider installs an enclosure (also known as a fiber distribution terminal (FDT)) on each floor, or every few floors, of an MDU. The FDT connects the building riser cable to the horizontal drop cables which run to each living unit on a floor. Drop cables are spliced or otherwise connected to the riser cable in the FDT only as service is requested from a tenant in a living unit. These service installations require multiple reentries to the enclosure, putting at risk the security and disruption of service to other tenants on the floor. This process also increases the service provider's capital and operating costs, as this type of connection requires the use of an expensive fusion splice machine and highly skilled labor. Routing and splicing individual drop cables can take an excessive amount of time, delaying the number of subscribers a technician can activate in one day, reducing revenues for the service provider. Alternatively, service providers install home run cabling the full extended length from each living unit in an MDU directly to a fiber distribution hub (FDH) in the building vault, therefore encompassing both the horizontal and riser with a single extended drop cable. This approach creates several challenges, including the necessity of first installing a pathway to manage, protect and hide each of the multiple drop cables. This pathway often includes very large (e.g., 2 inch to 4 inch to 6 inch) pre-fabricated crown molding made of wood, composite, or plastic. Many of these pathways, over time, become congested and disorganized, increasing the risk of service disruption due to fiber bends and excessive re-entry.

BRIEF SUMMARY OF THE INVENTION

[0005] Various embodiments described herein relate to fiber optic pathways, such as those used in various fiber optic applications. In general, the present disclosure relates to a system for routing optical fibers within multiple dwelling units (MDUs) where the system has a removable optical fiber feature that enables such routing of optical fibers within the MDUs.

[0006] In one embodiment, a method of installing a routing system and a jacketed optical fiber within a multiple dwelling unit (MDU), the routing system comprising a track having a channel configured to house the jacketed optical fiber is shown. The method comprising: adhering a portion of the track to a wall of the MDU; removing a portion of the jacketed optical fiber from the track of the routing structure; extending the jacketed optical fiber to create a slack loop; cutting the track to form a routing segment such that the routing segment is separated from the portion of the track that is adhered to the wall; and adhering the routing segment to the wall adjacent to the portion of the track that is adhered to the wall.

[0007] In another embodiment, adhering the routing segment to the wall comprises: forming a continuous channel between the portion of the track that is adhered to the wall and the routing segment. In another embodiment, the method further comprising: mounting a point of entry box to the routing system; and routing the jacketed optical fiber within the point of entry box. In another embodiment, the method, further comprising: mounting a point of entry box to the routing system;

and removing an outer jacket of the jacketed optical fiber to expose optical fibers within the jacketed optical fiber. In another embodiment, the method, further comprising: cutting an optical fiber housed in the jacketed optical fiber to form an optical fiber section; terminating the optical fiber section with a fiber optic connector; and connecting the fiber optic connector to an adapter within the point of entry box. In another embodiment, the track comprises an adhesive applied onto a surface of the track, the adhesive being covered by an adhesive strip. In another embodiment, adhering the track to the wall comprises: removing the adhesive strip; and applying the adhesive and the track onto the wall. In another embodiment, the jacketed optical fiber is pre-loaded into the channel of the track of the routing system. In another embodiment, the routing system further comprises a plurality of features, wherein each of the plurality of features are spaced apart along a length of the track by a gap. In another embodiment, a lifting tool is used to remove a portion of the jacketed optical fiber from the track by inserting a hook structure of the lifting tool into the gap between the features such that the hook structure engages with the portion of the jacketed optical fiber. In another embodiment, once the hook structure is engaged with the portion of the jacketed optical fiber, the portion of the jacketed optical fiber is lifted and extracted from the channel of the routing system. In another embodiment, adhering the track to the wall includes using an application tool to apply pressure onto the track such that the track adheres to the wall. In another embodiment, the cutting the track step further comprises: making a first cut and a second cut on the track, wherein the first cut and the second cut are spaced apart to create a track segment; wherein the track segment has a length that corresponds to the length of the slack loop. In another embodiment, the track segment is discarded prior to the adhering the routing segment step.

[0008] In one embodiment, a method of installing a routing system and a jacketed optical fiber along a corner of a wall within a multiple dwelling unit (MDU), the routing system comprising a track having a channel configured to house the jacketed optical fiber is provided. The method comprising: adhering a portion of the track to the wall of the MDU; removing a portion of the jacketed optical fiber from the track of the routing structure; cutting the track to form a routing segment such that the routing segment is separated from the portion of the track that is adhered to the wall; and adhering the routing segment to the wall along the corner of the wall of the MDU, wherein a corner of the routing segment is adjacent to a corner of the track.

[0009] In one embodiment, the track comprises an adhesive applied onto a surface of the track, the adhesive covered by an adhesive strip. In one embodiment, adhering the track to the wall comprises: removing the adhesive strip; and applying the adhesive and the track onto the wall. In one embodiment, the jacketed optical fiber is pre-loaded into the channel of the track of the routing system. In one embodiment, the routing system further comprises a plurality of features, wherein each of the plurality of features are spaced apart along a length of the track by a gap. In one embodiment, a lifting tool is used to remove a portion of the jacketed optical fiber from the track by inserting a hook structure of the lifting tool into the gap between the features such that the hook structure engages with the portion of the jacketed optical fiber. In one embodiment, once the hook structure is engaged with the portion of the jacketed optical fiber, the portion of the jacketed optical fiber is lifted and extracted from the channel of the routing system. In one embodiment, the cutting the track step further comprises: making a first cut and a second cut on the track, wherein the first cut and the second cut are spaced apart to create a track segment; wherein the track segment is discarded prior to the adhering the routing segment step.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale and are meant to be illustrative

and not limiting, and wherein:

[0011] FIG. **1** is a schematic view of an exemplary multiple dwelling units (MDU) having a drop access location system in accordance with the present disclosure;

[0012] FIG. **2A** is a schematic view of an exemplary final drop system in accordance with the present disclosure;

[0013] FIG. **2B** is a front view of a point of entry box that is used within the drop access system of FIG. **2A**;

[0014] FIG. **3A** is a perspective view of an optical fiber routing system in accordance with the present disclosure;

[0015] FIG. **3B** is a perspective view of the optical fiber routing system of FIG. **3A** with an optical fiber cable housed within the optical fiber routing system;

[0016] FIGS. **4-11** illustrate a method of installing the routing system within an MDU in accordance with the present disclosure; and

[0017] FIGS. **12-14** illustrate a method of installing the routing system to account for a corner on a wall within an MDU.

DETAILED DESCRIPTION OF THE INVENTION

[0018] The present invention now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

[0019] In general, the present disclosure relates to a system for routing optical fibers within multiple dwelling units (MDUs) where the system has a removable optical fiber feature that enables such routing of optical fibers within the MDUs.

[0020] Referring first to FIG. **1**, an example MDU **100** that can accommodate the exemplary drop access system described herein is shown. MDU **100** is a multi-floor structure having a plurality of living units **103** located therein. One example floor **102** has four living units **103** having a common hallway **104**. Feeder cable **10** brings communications lines (hereinafter referred to as “optical fiber cables”) to and from building **100**. These feeder lines are spliced to the MDU's cabling at a splice closure **20**. The building feeder lines **30** are distributed to the building from a fiber distribution hub (FDH) **40**. Each floor includes a fiber distribution terminal (FDT) **50** that receives optical fiber cables via riser cable **55**. In the present example, a drop access system **101** coupling the optical fiber cables from FDT **50a** can be installed on hallway **104** where drop access boxes **150** can be disposed at each living unit and can receive one or more optical fiber cables from duct **110**.

[0021] As is also mentioned above, the drop access system **101** can be configured to accommodate electrical wire drops and hybrid combination drops as well. In alternative aspects, the drop access system **101** can be configured to supply at least one of uninterrupted DC power and AC power to an optical network terminal located in an individual living unit at the drop location.

[0022] Referring now to FIG. **2A**, a schematic view of a drop access system **101** is shown. System **101** is installed in a hallway of an exemplary MDU **100**. System **101** can be utilized in other indoor and outdoor applications, and in commercial or residential buildings, such as in office buildings, professional suites, and apartment buildings. The drop access system **101** comprises a conduit or duct or track **202** which contains one or more jacketed optical fibers **210** (FIG. **3B**) from a telecommunications closet (or other distribution location) to one or more living units **103**. The jacketed optical fibers **210** can comprise optical fibers for data, video, and/or telephone signal transmission. In one aspect, the jacketed optical fibers **210** can comprise discrete (loose) or ribbonized optical fiber, such as 900 μm buffered optical fiber(s) or other standard size communications optical fiber. In addition, in another aspect, the jacketed optical fibers **210** can comprise optical fibers **210** that can be single mode or multimode optical fibers. While the jacketed

optical fiber **210** shows a single optical fiber **210A** in FIG. 3B, it is within the scope of the present disclosure that jacketed optical fiber **210** can comprise multiple optical fibers **210A**. In addition, although the exemplary aspects described herein are often specific to accessing optical fiber lines, it would be understood by one of ordinary skill in the art given the present description that the system can be configured to accommodate electrical wire drops and hybrid combination drops (optical fibers and electrical wires) as well.

[0023] System **101** can comprise one or more point of entry boxes **150** that are located at one or more access points **105**, such as at or near the entryway of a living unit. As is described in U.S. Pat. No. 8,107,785, incorporated by reference herein in its entirety, an exemplary point of entry box **150** can include a base portion and a cover, where the access box (cover and base) can have a low profile and/or decorative outer design. Alternatively, the point of entry box can be designed as a low profile base unit for a drop fiber point of entry system, such as is described in the pending U.S. Pat. No. 8,295,670, incorporated by reference herein in its entirety. In addition, the system **101** can be installed in the hallway or passageway of an MDU as is described in U.S. Pat. No. 8,107,785, incorporated by reference above.

[0024] In addition and as shown in FIGS. 2A and 2B, the point of entry box **150** can include a mounting section **152** that provides for straightforward mounting of the point of entry box **150** onto the track **202**. Mounting section **152** is configured to fit onto and over duct **110**. In this manner, point of entry box **150** can be mounted to track **202** after the track **202** (and jacketed optical fibers **210** therein) are already installed. In addition, the configuration of mounting section **152** allows for point of entry box **150** to be mounted onto track **202** at nearly any location along the track path. This configuration allows a through hole to be drilled into the living unit **103** at a later time. In addition, the configuration of system **101** allows for multiple possible access box locations, including above door locations or at lower height locations in the hallway **104** of the MDU **100**. Furthermore, point of entry box **150** includes an adapter **154** that is configured to couple at least one jacketed optical fiber **210** or optical fiber **210A** to a subscriber line within the living unit **103** as discussed in greater detail herein. The wall mounting portion **152** further includes routing structures **156** to provide a slack storage section for storing excess amounts of the jacketed optical fiber **210** or optical fibers **210A** housed within jacketed optical fiber **210** that are fed through track **202**. For example, a series of tabs or other structures disposed in a recess of the wall mounting portion **152** can be configured to loosely secure excess optical fibers **210A**. The optical fibers **210A** can be protected from over-bending by configuring the wall mounting portion **152** to have a suitable radius or width.

[0025] Referring now to FIGS. 3A and 3B, an optical fiber routing system (hereinafter referred to as “routing system”) **200** is shown. Routing system **200** comprises a track **202** and features **204** that are positioned along its length. As shown, in some embodiments, the features **204** are positioned on opposing sides of the channel **206**. In some embodiments, the features **204** may be intermittently spaced along the length of the track. In one embodiment, the features **204** can be post-like features that may include a widened cap at the top of the features (much like mushrooms), such as those illustrated in FIGS. 3A and 3B, where the cap aids in securing and protecting the jacketed optical fiber **210** in the channel **206**. The jacketed optical fiber **210** may be inserted into the channel **206** using an insertion tool that applies pressure onto the jacketed optical fiber **210** in the direction of the channel **206** in order to slide the jacketed optical fiber **210** past the any cap or top of the features and secure the jacketed optical fiber **210** in the channel **206**.

[0026] The surface of the track that is intended for adhesion to the wall, i.e., surface **208**, may be pre-laminated with some sort of adhesive layer **209**, e.g., a pressure sensitive adhesive layer. The adhesive may be a double sided adhesive for sticking to both the track and to the interior wall of a dwelling. The side of the adhesive intended for adhesion to the interior wall may initially be covered with a liner **212** (shown, but optional), that may be stripped away by the installer at the time of application of the track to the wall. In one embodiment, the adhesive layer **209** may be a

stretch-release adhesive. Any appropriate type of adhesive for use in adhesive layer **209** is contemplated.

[0027] In some embodiments, the track **202** and the features **204** will be made up of a flexible material that is either clear or translucent. The flexibility of the track **202** and features **204** enable the track **202** to bend en route to the wall outlet, where, e.g., the track must turn a corner. Additionally, flexibility of the features **204** allows for the optical fiber cable **210** to be snapped into the channel **206**. The clear or translucent nature of the track **202** allows it to be aesthetically pleasing and difficult to notice for tenants or workers in MDU **100**. Alternatively, the track **202** and features **204** may be made up of a flexible material that is color-matched to the color of the wall onto which it is being adhered. Similarly, in order to make jacketed optical fiber **210** less conspicuous, the buffer coating around the optical fiber **210A** routed in the track **202** may be either clear or translucent or may be color-matched to the color of the wall onto which the track **202** is adhered. Color-matching the track or buffered optical fiber **210A** may offer similar concealing, aesthetically pleasing properties to the system.

[0028] In some embodiments, a cover portion may be applied over the channel **206** as described in U.S. Pat. No. 10,018,798, which is incorporated by reference herein in its entirety.

[0029] Referring now to FIGS. **4-11**, a method of installing routing system **200** within drop access system **101** is shown. Referring first to FIG. **4**, routing system **200** is installed onto a wall **201** within MDU **100**. As mentioned previously, the wall **201** may be in a hallway of MDU **100**. However, it is within the scope of the present disclosure that the wall **201** may be within a living unit **103** of MDU **100**. When installing track **202** on the wall **201**, liner **212** is removed such that adhesive layer **209** is exposed and applied onto the wall to adhere track **202** to the wall **201**. In some embodiments, a jacketed optical fiber **210** or optical fiber **210A** is pre-loaded into channel **206** of track **202**, and an application tool **207** is used along the length of track **202** to apply pressure onto the routing system **200** and further secure track **202** to wall **201** and jacketed optical fiber **210** or optical fiber **210A** within channel **206** of track **202**. In particular, jacketed optical fiber **210** or optical fiber **210A** is installed into channel **206** of track **202** when routing system **200** is manufactured in the factory, and the routing system **200** with the pre-installed jacketed optical fiber **210** or optical fiber **210A** is provided on a spool or reel from which the routing system **200** is drawn and applied onto wall **201** as described above. In other embodiments, a jacketed optical fiber cable **210** or optical fiber **210A** is loaded into channel **206** of track **202**, and as shown, an application tool **207** is used along the length of track **202** to apply pressure onto the track **202** and onto optical fiber **210A** such that the track **202** adheres to the wall **201** and the optical fiber **210A** is secured within channel **206** of track **202**. While an application tool **207** is described in the present disclosure, it is within the scope of the present disclosure that alternate methods of applying the track and/or optical fiber **210A** onto wall **201** may be used (e.g., a technician's hands, etc.).

[0030] Then, with continued reference to FIG. **4**, a portion of the jacketed optical fiber **210** that is seated within track **202** is removed. To remove the portion of jacketed optical fiber **210**, a lifting tool **205** can be used to lift the portion of the jacketed optical fiber **210** from the channel **206**. In particular, lifting tool **205** includes a hook structure **205A** that can be inserted between features **204** within gap **204A** such that hook structure **205A** can engage with jacketed optical fiber **210**. Once hook structure **205A** engages with the jacketed optical fiber **210**, lifting tool **205** is lifted such that hook structure **205A** lifts the portion of the jacketed optical fiber **210** from the channel **206** of track **202** and beyond features **204**. Once the portion of the jacketed optical fiber **210** is removed, the jacketed optical fiber **210** is pulled or actuated such that a desired length of the jacketed optical fiber **210** is extracted from track **202** for creating a slack loop as discussed below. For example, if a desired length of thirty six (36) inches of jacketed optical fiber **210** is needed, then thirty-six (36) inches of jacketed optical fiber **210** is extracted from track **202** downstream of the initial extraction point to create a slack loop having that length of jacketed optical fiber **210**.

[0031] After the desired length of the jacketed optical fiber **210** is removed, track **202** of routing

system **200** is cut where the jacketed optical fiber **210** was removed as shown in FIG. 5 to form a routing segment **200'**. In particular, track **202** is cut in two places that correspond to the length of the jacketed optical fiber **210** that is removed from track **202** as shown. The two cuts as shown create a track segment **202'** that is removed as shown in FIG. 5 and discussed below.

[0032] Once the slack loop is created as shown in FIG. 6, track segment **202'** is removed or discarded and routing segment **200'** is installed onto the wall **201** such that routing segment **200'** and routing system **200** abut one another and are aligned to re-establish the continuity of channel **206**. That is, channel **206** of routing segment **200'** and channel **206** of routing system **200** are aligned to form a continuous channel **206** for the jacketed optical fiber **210**. Routing segment **200'** is installed onto the wall **201** similar to how routing system **200** is installed-by removing liner **212** and applying adhesive layer **209** onto the wall **201**.

[0033] Then, as shown in FIG. 7, the wall mounting portion **152** of drop access box **150** is mounted on the wall **201** such that the slack loop is integrated into the drop access box **150** as described herein.

[0034] In one embodiment, as shown in FIGS. 7A and 7B, the slack loop of the jacketed optical fiber **210** is routed around routing structures **156** to be encased in the wall mounting portion **152** of drop access box **150**.

[0035] In another embodiment, as shown in FIGS. 8-11, one of the optical fibers **210A** is terminated such that the optical fiber **210A** is coupled to the adapter **154** and connects a subscriber line (i.e., living unit **103**) to the corresponding FDT **50** (FIG. 1) as discussed below. In particular, first, as shown in FIG. 8, a cutting tool **213** is used to cut and remove the covering of optical fiber cable **210** and expose the optical fibers **210A** within the optical fiber cable **210**. Then, as shown in FIG. 9, a single optical fiber **210A** within the group of optical fibers **210A** is cleaved with a cutting tool (e.g., scissors) to create an optical fiber section **210A'** as shown in FIG. 10. As shown, the optical fiber section **210A'** is terminated with a fiber optic connector **211** by methods that are known in the art. Finally, the fiber optic connector **211** of the terminated optical fiber section is then coupled to adapter **154** of drop access box **150** as shown in FIG. 11.

[0036] Referring now to FIGS. 12-14, a method of installing the routing system **200** along a corner of a wall **201** is shown. Referring first to FIG. 12, routing system **200** is installed onto a wall **201** within MDU **100**. As mentioned previously, the wall **201** may be in a hallway of MDU **100**. However, it is within the scope of the present disclosure that the wall **201** may be within a living unit **103** of MDU **100**. When installing track **202** on the wall **201**, liner **212** is removed such that adhesive layer **209** is exposed and applied onto the wall to adhere track **202** to the wall **201**. In some embodiments, a jacketed optical fiber **210** or optical fiber **210A** is pre-loaded into channel **206** of track **202**, and an application tool **207** is used along the length of track **202** to apply pressure onto the routing system **200** and further secure track **202** to wall **201** and optical fiber **210A** within channel **206** of track **202**. In other embodiments, a jacketed optical fiber cable **210** or optical fiber **210A** is loaded into channel **206** of track **202**, and as shown, an application tool **207** is used along the length of track **202** to apply pressure onto the track **202** and onto optical fiber **210A** such that the track **202** adheres to the wall **201** and the optical fiber **210A** is secured within channel **206** of track **202**. While an application tool **207** is described in the present disclosure, it is within the scope of the present disclosure that alternate methods of applying the track and/or optical fiber **210A** onto wall **201** may be used (e.g., a technician's hands, etc.).

[0037] Then, with continued reference to FIG. 12, a portion of the jacketed optical fiber **210** that is seated within track **202** is removed. To remove the portion of jacketed optical fiber **210**, lifting tool **205** can be used to lift the portion of the jacketed optical fiber **210** from the channel **206**. In particular, lifting tool **205** includes a hook structure **205A** that can be inserted between features **204** within gap **204A** such that hook structure **205A** can engage with jacketed optical fiber **210**. Once hook structure **205A** engages with the jacketed optical fiber **210**, lifting tool **205** is lifted such that hook structure **205A** lifts the portion of the jacketed optical fiber **210** from the channel **206** of track

202 and beyond features **204**. Once the portion of the jacketed optical fiber **210** is removed, the jacketed optical fiber **210** is pulled or actuated such that a desired length of the jacketed optical fiber **210** is extracted from track **202** as discussed below. In one embodiment, as shown, jacketed optical fiber **210** is actuated such that channel **206** is empty for the corresponding length spanning two (2) features **204** (FIGS. 3A and 3B). While the present disclosure discloses jacketed optical fibers **210** being removed for the corresponding length of two features **204**, it is within the scope of the present disclosure that alternate lengths of jacketed optical fibers **210** may be removed from track **202** by this process.

[0038] After the desired length of the jacketed optical fiber **210** is removed, track **202** of routing system **200** is cut where the jacketed optical fiber **210** was removed as shown in FIG. 13 to form a routing segment **200'**. In particular, track **202** is cut in two places that correspond to the length of the jacketed optical fiber **210** that is removed from track **202** as shown. The two cuts as shown create a track segment **202'** that is removed as shown in FIG. 13 and discussed below. By cutting and removing track segment **202'**, the length of the jacketed optical fiber **210** has an increased length relative to the track **202**, which enables bending of the jacketed optical fiber **210** as discussed in detail below.

[0039] Once track segment **202'** is removed or discarded, routing segment **200'** is installed onto the wall **201** as shown in FIG. 14. In particular, routing segment **200'** is installed onto the wall **201** such that a corner **214** of the routing segment **200'** is in contact with a corner **215** of the track **202**. Routing segment **200'** is installed onto the wall **201** at an angle relative to routing system **200**. Routing segment **200'** is installed similar to how routing system **200** is installed-by removing liner **212** and applying adhesive layer **209** onto the wall **201**. This configuration enables jacketed optical fiber **210** to bend along the corner of the wall **201** without impacting the optical fibers **210A** that are housed within jacketed optical fiber **210**.

[0040] While bending a jacketed optical fiber **210** along a corner of wall **201** is disclosed, it is within the scope of the present disclosure that the method disclosed in FIGS. 12-14 enables jacketed optical fiber to be directed around various objects that could be in the pathway of routing system **200**. For example, a sign may be mounted onto wall **201**, and the method of FIGS. 12-14 enables routing system **200** and jacketed optical fiber **210** to be directed around the sign without the optical fibers **210A** within jacketed optical fiber **210** incurring bend loss. Moreover, as shown, the method disclosed in FIGS. 12-14 enables a ninety (90) degree transition within a single plane of a room as discussed above.

[0041] Many modifications and other embodiments of the invention set forth herein will come to mind to one skilled in the art to which the invention pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the invention is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. It is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A method of installing a routing system and a jacketed optical fiber within a multiple dwelling unit (MDU), the routing system comprising a track having a channel configured to house the jacketed optical fiber, the method comprising: adhering a portion of the track to a wall of the MDU; removing a portion of the jacketed optical fiber from the track of the routing structure; extending the jacketed optical fiber to create a slack loop; cutting the track to form a routing segment such that the routing segment is separated from the portion of the track that is adhered to the wall; and

adhering the routing segment to the wall adjacent to the portion of the track that is adhered to the wall.

2. The method of claim 1, wherein adhering the routing segment to the wall comprises: forming a continuous channel between the portion of the track that is adhered to the wall and the routing segment.

3. The method of claim 1, further comprising: mounting a point of entry box to the routing system; and routing the jacketed optical fiber within the point of entry box.

4. The method of claim 1, further comprising: mounting a point of entry box to the routing system; and removing an outer jacket of the jacketed optical fiber to expose optical fibers within the jacketed optical fiber.

5. The method of claim 1, further comprising: cutting an optical fiber housed in the jacketed optical fiber to form an optical fiber section; terminating the optical fiber section with a fiber optic connector; and connecting the fiber optic connector to an adapter within the point of entry box.

6. The method of claim 1, wherein the track comprises an adhesive applied onto a surface of the track, the adhesive being covered by an adhesive strip.

7. The method of claim 6, wherein adhering the track to the wall comprises: removing the adhesive strip; and applying the adhesive and the track onto the wall.

8. The method of claim 1, wherein the jacketed optical fiber is pre-loaded into the channel of the track of the routing system.

9. The method of claim 1, wherein the routing system further comprises a plurality of features, wherein each of the plurality of features are spaced apart along a length of the track by a gap.

10. The method of claim 9, wherein a lifting tool is used to remove a portion of the jacketed optical fiber from the track by inserting a hook structure of the lifting tool into the gap between the features such that the hook structure engages with the portion of the jacketed optical fiber.

11. The method of claim 10, wherein once the hook structure is engaged with the portion of the jacketed optical fiber, the portion of the jacketed optical fiber is lifted and extracted from the channel of the routing system.

12. The method of claim 1, wherein adhering the track to the wall includes using an application tool to apply pressure onto the track such that the track adheres to the wall.

13. The method of claim 1, wherein the cutting the track step further comprises: making a first cut and a second cut on the track, wherein the first cut and the second cut are spaced apart to create a track segment; wherein the track segment has a length that corresponds to the length of the slack loop.

14. The method of claim 13, wherein the track segment is discarded prior to the adhering the routing segment step.

15. A method of installing a routing system and a jacketed optical fiber along a corner of a wall within a multiple dwelling unit (MDU), the routing system comprising a track having a channel configured to house the jacketed optical fiber, the method comprising: adhering a portion of the track to the wall of the MDU; removing a portion of the jacketed optical fiber from the track of the routing structure; cutting the track to form a routing segment such that the routing segment is separated from the portion of the track that is adhered to the wall; and adhering the routing segment to the wall along the corner of the wall of the MDU, wherein a corner of the routing segment is adjacent to a corner of the track.

16. The method of claim 15, wherein the track comprises an adhesive applied onto a surface of the track, the adhesive covered by an adhesive strip.

17. The method of claim 16, wherein adhering the track to the wall comprises: removing the adhesive strip; and applying the adhesive and the track onto the wall.

18. The method of claim 15, wherein the jacketed optical fiber is pre-loaded into the channel of the track of the routing system.

19. The method of claim 15, wherein the routing system further comprises a plurality of features,

wherein each of the plurality of features are spaced apart along a length of the track by a gap.

20. The method of claim 19, wherein a lifting tool is used to remove a portion of the jacketed optical fiber from the track by inserting a hook structure of the lifting tool into the gap between the features such that the hook structure engages with the portion of the jacketed optical fiber.

21. The method of claim 20, wherein once the hook structure is engaged with the portion of the jacketed optical fiber, the portion of the jacketed optical fiber is lifted and extracted from the channel of the routing system.

22. The method of claim 15, wherein the cutting the track step further comprises: making a first cut and a second cut on the track, wherein the first cut and the second cut are spaced apart to create a track segment; wherein the track segment is discarded prior to the adhering the routing segment step.
